

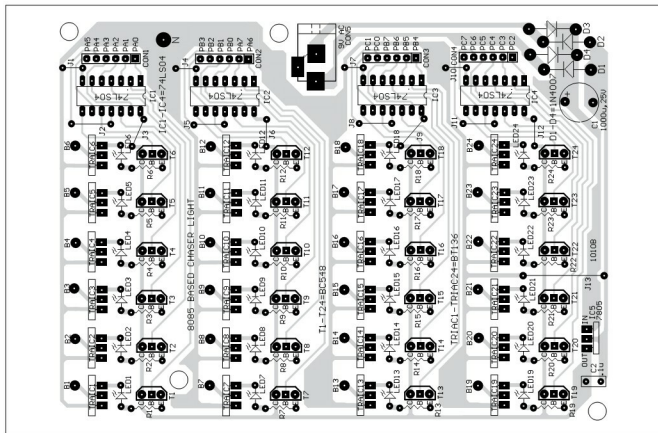


Fig. 4: Component layout for the PCB

key of the kit to clear the display. Now enter the light pattern number, say, '1'. P1 appears on the display and pattern-1 starts running. Similarly, other light patterns (pattern 2 through 5) can be run.

For addressing the peripheral device (8255, 8279), input/output (I/O) mapped scheme is used. The addresses have been modified in accordance with the Dynalog kit used for the

purpose. You would need to modify the program address space as well as input/output address for the peripheral suitably, in accordance with the specific kit used by you. Modifications due to the kit are marked with asterisk (*) and these can be changed accordingly.

Caution. Since switching is accomplished by triacs, don't touch the circuit parts while AC supply

is plugged in. Further, ensure that Live and Neutral lines are not interchanged.

EFY note. The source code of this project has been included in this month's EFY-CD and is also available on www.efymag.com website.

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CHASER.LST

;Chaser Light Program Environment

```
;Stack pointer initialised at : 91FFH
;Port-A (output) of 8255 : 08H
;Port-B (output) of 8255 : 09H
;Port-C (output) of 8255 : 0AH
;Control word of 8255 : 0BH
;Control word of 8279 : 01H
;Data address of 8279 : 00H
;RDKBD subroutine of kit location: 0A64H
;CLEAR subroutine of kit location: 0B00H
;Value of 'Next' key of kit : 11H
```

Program

```
0001 0000 ;PROGRAM ENVIRONMENT
0001 0000 ;PROGRAM ENVIRONMENT
```

```
0002 0000 ;RAM locations used for the program
; 9000-9FFF
0003 0000 STACK .EQU 91FFH* ;Stack pointer
; initialised at 91FF
0004 0000 PORTA .EQU 08H* ;PORT A of 8255
0005 0000 PORTB .EQU 09H* ;PORT B of 8255
0006 0000 PORTC .EQU 0AH* ;PORT C of 8255
0007 0000 CWR (8255) .EQU 0BH* ;Control Word of 8255
0008 0000 DAT (8279) .EQU 01H* ;Control Word of 8279
0009 0000 DAT (8279) .EQU 00H* ;Data Address of 8279
0010 0000 RDKBD .EQU 0A64H* ;Location of RIT
; keyboard subroutine
0011 0000 CLEAR .EQU 0B00H* ;Location of display
; Clear subroutine
0012 0000 .ORG 9000H
```